

Challenges regarding implementation of electronic chain traceability

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Abstract

Purpose – Increased focus on safe and healthy food has resulted in the need for implementing electronic chain traceability. This poses challenges both regarding technical solutions and business processes. The purpose of this paper is to report from a research project where implementation in a value chain for farmed salmon is used to investigate these challenges.

Design/methodology/approach – The first step in the project was to establish a complete value chain for farmed salmon as a basis for further study. The second step was to use theoretical concepts of internal traceability and chain traceability as a basis for analyzing the status of each company regarding readiness to implement electronic chain traceability. The third step was to compare the final level of implementation in each company with the level of implementation readiness at the start of the project.

Findings – Implementation of electronic chain traceability may involve changes both in work processes and software systems. This study shows that a set of criteria regarding implementation readiness can be used to assess the challenges and risks involved. Coherence between implementation readiness and the final level of implementation is indicated.

Research limitations/implications – This research is limited to companies in a specific food sector. The generalization of the results for use in other sectors needs to be investigated.

Practical implications – Because of the increasing demand for improving chain traceability, it can be expected that implementation readiness regarding electronic chain traceability solutions will be an important factor in selecting partners in food value chains.

Originality/value – The large number of companies involved in this study is a good basis for evaluating the complexity of implementing electronic chain traceability. This paper describes guidelines and assessment criteria for companies planning to implement such solutions.

Keywords Food industry, Value chain, Fish farming, Electronic data interchange

Paper type Research paper

Introduction

Food scandals and product recalls are some of the driving forces behind the more stringent traceability legislation and regulations we have seen the last few years, both in Europe and the US. The European Union (EU) “General Food Law Regulation” (EU, 2002) came into effect in January 2005, while the US Bioterrorism act (US FDA, 2002) has been in effect from December 2003. Many companies have implemented internal traceability systems. This is not mandatory in order to comply with the General Food

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Law Regulations, but as the Food Standard Agency in the UK states “IT based traceability systems are regarded as more effective compared to paper based systems” (FSA, 2002).

Analysis of existing traceability systems shows that only a few links in a supply chain are using software for internal traceability. The diversity of the systems also makes the integration difficult (Bechini *et al.*, 2005). The preliminary report of the Food Standards Agency (FSA, 2002) stated correspondingly that a better integration of electronic data exchange is needed.

An international standard, often referred to as “the TraceFish standard” describes the principles of traceability for the fish farming and fishing industries (CEN, 2003a, b). An investigation of traceability of fish products in Iceland and China concluded that the TraceFish standard was helpful to companies who are aiming at implementing traceability systems for their products (Liu and Olafsdottir, 2002). This standard was used as the basis for establishing an XML standard for electronic exchange of traceability information between links in a chain (www.TraceFish.org). Subsequently there have been developed multiple generic models for utilizing XML for traceability information exchange (Lo Bello *et al.*, 2004), (Bechini *et al.*, 2005), (Folinas *et al.*, 2006). These models are mainly divided into two groups; distributed or centralized architecture (Bechini *et al.*, 2005) depending on where the traceability information is stored.

This paper describes the challenges and experiences related to the implementation of a centralized electronic chain traceability solution in a complete value chain for farmed salmon – from feed additives to salmon ready for consumption.

The first part of the paper gives a general background on traceability and the basic requirements for internal traceability and chain traceability. The importance of unique and standardized identification of traceable units is explained.

The implementation study with the demonstration of an electronic solution for a chain traceability pilot with eight companies is described in detail. The analysis of business processes and recommendations for changes in each company are the basis for the implementation of new processes and technical solutions.

The emphasis in the paper is on the challenges and experiences encountered during the analysis and implementation. As a basis for the evaluation, specific criteria were defined for use in an implementation readiness assessment with focus on implementing electronic chain traceability. The software concept used in the implementation is based on a net-centric solution, as opposed to point-to-point transfer of data between companies in the value chain. Different aspects regarding this solution, as well as future directions, are discussed.

Traceability principles

The general Food Law (EU, 2002) requires chain traceability systems. The guidance on the implementation of EC Food Law Regulation Article 18 (Guide 178/2002) declare that “it is in the logic of Article 18 that a certain level of internal traceability would be put in place by food business operators”. One of the basic prerequisites of both internal and chain traceability is the unique identification of raw materials, semifinished products and finished products. This is also pointed out by Bechini *et al.* (2005). A unique identification must be linked to a clearly defined set of raw materials or

products. The term “traceable unit” is used as a common denotation for such a set in this article.

It is usually an advantage that the contents of a traceable unit are homogeneous and have the same product properties. This could for example be necessary for follow-up on production processes, statistical verification procedures, issuing quality certificates etc.

The principles of chain traceability and internal traceability are shown in Figure 1. The processes in each company are based on the CSCMP (Council of Supply Chain Management Professionals) definitions (CSCMP, 2003).

As a basis for chain traceability, the identities for traceable units must be recorded at reception and shipping (see top vertical arrows in Figure 1). Internally in each link in the chain, raw materials are processed in one or more steps to finished products. Note also that the identities for both external and internal returns should be recorded.

A simple example is shown in Figure 2. Company C sources one unit from company A and two from company B. These units are used as input for producing two units delivered to company D and one to company E. These traceable units are all identified by a globally unique SGTIN – Serialized Global Trade Item Number (EPCglobal, 2006). The SGTIN is designed for encoding in an RFID tag (Finkenzeller, 2003), but the principle of unique identification can be achieved using other methods (e.g. barcodes, two-dimensional barcodes, even manual data entry).

Company C is one of the links in a chain and needs to record the unique identities of all sourced and delivered units (EU, 2002). In addition to this, links between sourced and delivered units need to be established. These links can be shown by a simplified data model (see Figure 3). For most production companies the relation between sourced and delivered units is a many-to-many relation.

Figure 4 shows this data model used on the product flow in Figure 2, with a very simple production process where all the delivered units contain parts of two sourced units. The arrows show how the contents of a delivered unit can be traced through the relationship table *IngredientsInDelivered*.

This example illustrates that internal traceability requires recording of all transformations during the production processes. This implies that also intermediate products need unique identification. Typical internal data capture points are shown

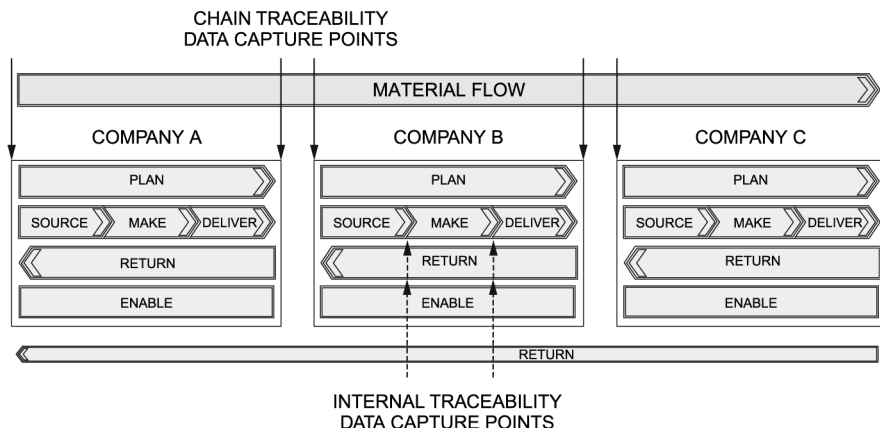


Figure 1.
Location of traceability
data capture points

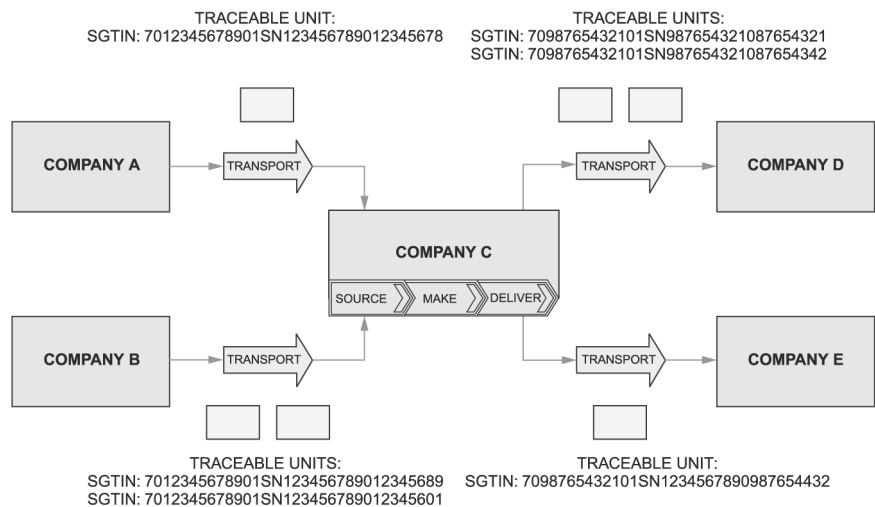


Figure 2.
Example, tracing of
sourcing and delivery



Figure 3.
Transformation from
sourced to delivered units

TABLE: SGTIN_Sourced	
SGTIN_Sourced	Attributes 1..N
7012345678901SN123456789012345601	
7012345678901SN123456789012345678	
7012345678901SN123456789012345689	

TABLE: IngredientsInDelivered		
SGTIN_Sourced	SGTIN_Delivered	Attributes 1..N
7012345678901SN123456789012345601	7098765432101SN987654321087654321	
7012345678901SN123456789012345601	7098765432101SN987654321087654342	
7012345678901SN123456789012345678	7098765432101SN987654321087654321	
7012345678901SN123456789012345678	7098765432101SN987654321087654321	
7012345678901SN123456789012345689	7098765432101SN987654321087654342	
7012345678901SN123456789012345689	7098765432101SN987654321087654342	

TABLE: SGTIN_Delivered	
SGTIN_Delivered	Attributes 1..N
7098765432101SN987654321087654321	
7098765432101SN987654321087654342	
7098765432101SN987654321087654342	

Figure 4.
Relational database
implementation principle

more in detail in Figure 5. The identification and data capture technologies used will vary depending on the production process, product properties, technology maturity in the company etc. The data collected from each process can be the input to an internal traceability system, which links sourced units to the corresponding delivered units.

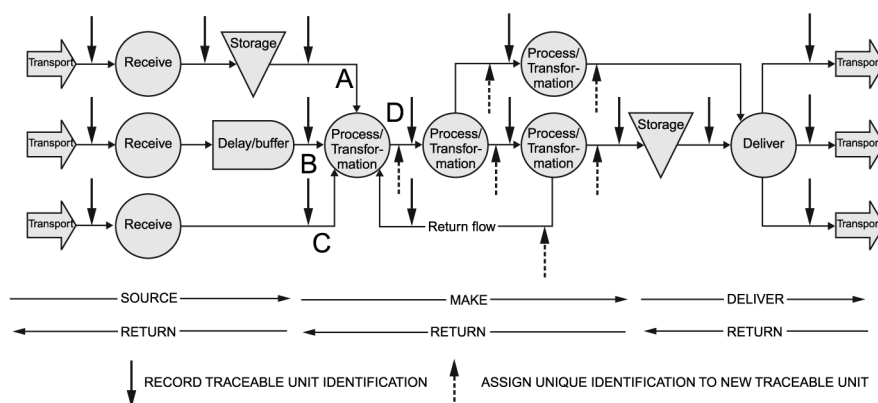


Figure 5.
Detailed view of internal
data capture points

Basically, three types of operations are necessary for obtaining internal traceability:

- (1) Record the unique identities of traceable units (typically input to processes).
- (2) Assign unique identities to new traceable units (typically output from processes).
- (3) Link a set of input (sourced) unit identities to one or more sets of output (delivered) identities (this typically represents a transformation of raw materials to finished products).

Note that without internal traceability, all delivered units must be assumed to contain elements from all sourced units. This leads to a dramatic increase in possible combinations.

Principles for internal and chain traceability in the distribution chains for farmed and captured fish have been defined in two CEN standards called TraceFish (CEN, 2003a, b). The TraceFish standards do not specify methods of identification or data capture, but it is obvious that increasingly complex production processes and value chains require electronic data capture and data exchange. A preliminary report made by the Food Standard Agency in the UK also concludes with the fact that “IT based traceability systems are required as more effective compared to paper based systems” (FSA, 2002). This was the main reason for studying the challenges related to implementing a software concept for electronic chain traceability. This implementation study is described in the next section. TraceFish was used as the starting point for analysis in all companies participating in the study.

The chain traceability implementation study

Scope of the TELOP Trace project

One of the biggest challenges of food chain traceability is the extensive use of manual exchange of information between companies. The results from surveys made in a joint Nordic project (Egilsson *et al.*, 1999) and background material for the TraceFish standards (unpublished) show that manual systems for information exchange between companies are predominating in the fish farming industry. This finding is corresponding with other surveys (Bechini *et al.*, 2005). Manual recording methods are regarded as labour-intensive and error prone. Electronic data exchange of

traceability data for all ingredients is expected to reduce the workload and increase the accuracy of the recording concerning all input factors in the food production. At the same time the diversity of internal traceability software makes integration difficult (Bechini *et al.*, 2005). The main objective in the TELOP Trace project (2003-2005) was to demonstrate a software concept for electronic chain traceability in a complete value chain.

The TELOP Trace project chain is shown in Figure 6. The chain consists of eight different companies plus adjacent transporters. The majority of the companies included in the chain were defined at the project start. Two of the companies joined the project at a later stage. Only one of the multiple transport links was included in the implementation.

The project guidelines required that the companies are anonymous in public material, but they span from large international industrial groups to small local companies. Locations in three European countries were involved in the study.

Methodology used in the chain traceability survey

The first step in the project was to establish a complete value chain for farmed salmon as a basis for further study. This included defining requirements for the electronic chain traceability solution. The second step was to use theoretical concepts of internal traceability and chain traceability as a basis for analysing the status of each company regarding readiness to implement electronic chain traceability. The third step was to compare the final level of implementation in each company with the level of implementation readiness at the start of the project.

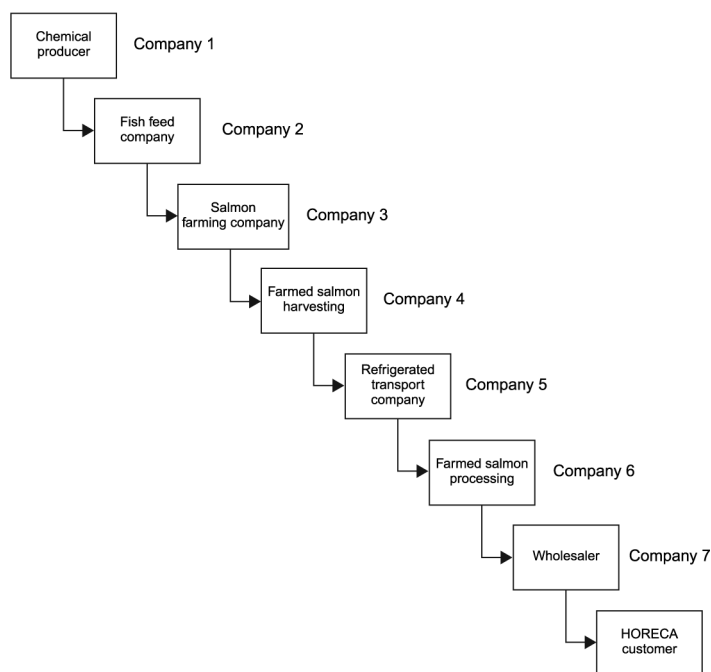


Figure 6.
The TELOP trace chain

The different companies in the project chain had various statuses regarding readiness to implement electronic chain traceability. To define the level of readiness a survey was carried out. The main focus of the survey was to:

- identify internal and chain traceability data capture points at each link in the chain;
 - document what traceability information was recorded at internal and chain traceability data capture points (Figure 1) at each link in the chain;
 - document how the recorded information was stored; and
 - document how and what traceability information was transferred between the links in the chain.
-

Surveys were based on site visits and interviews with key personnel at each company. Individual reports for each company were made.

The following criteria have been identified as essential (Forås *et al.*, 2004) for the implementation of electronic chain traceability in a company:

- Presence of chain traceability procedures at point of reception: Routines regarding recording of traceability information for all received input factors (not relevant for the first link in a traceability chain).
- Use of electronic/automatic data recording technology at reception.
- Presence of internal traceability procedures: routines regarding recording of process information and transformation information. The evaluation included all input factors.
- Presence of an internal traceability software system.
- Use of standard ID on trade units: GS1 numbering schemes; SSCC, GTIN, SGTIN (GS1, 2006).
- Use of electronic/automatic data recording technology at dispatch.
- Presence of chain traceability procedures at point of dispatch: routines regarding recording of traceability information for all dispatched trade units.
- Presence of chain traceability software system: use of electronic exchange of traceability information to customers.

All eight Implementation Readiness Criteria (IRC) in this project are regarded as of equal importance for the purpose of estimating implementation readiness. All surveys were analysed to detect the total number of IRC present in each company. The final level of implementation was also evaluated at the end of the project period, and quantified according to the following criteria:

- 1 = Online electronic implementation with information sent bidirectional between local ERP systems and the chain traceability system.
- 0.75 = electronic implementation with information sent unidirectional from local ERP systems to the chain traceability system.
- 0.5 = Transfer of files with extracts from local ERP databases to the chain traceability system.

- 0.25 = Transfer of files with extracts from internal paper files to the chain traceability system.
- 0 = No transfer of traceability information to the chain traceability system.

Software concept and implementation

The project determined the following requirements for the electronic chain traceability solution:

- information at the fingertips of the user, with an easy and intuitive navigation;
- a global solution, applicable in any geography/language;
- independence from ERP and internal systems;
- a Generic Traceability Model that could be applied and adapted to any type of business (production, retail, distribution, transport) or industry;
- data fully controlled by data owner;
- ease of connection and use; and
- security.

In order to solve these requirements, three main architectures were analysed for the value chain: point to point connections, the use of an external central database and a net-centric service. Figure 7 compares the advantages and disadvantages of these three alternatives.

It was decided to base the technical platform for the TELOP project on a net-centric service. The service allows for connection to different systems as shown in Figure 8, and gathers relevant traceability information in the value chain.

In order to provide chain traceability, the service requires that internal traceability systems are in place. For technical reasons it was necessary to provide a local system as interface between local ERP systems and the net-centric solution for the project period (TTES-TraceTracker Enterprise Solution, see Figure 8).

Data to TTES was imported from different local platforms, both general purpose ERP systems and sector specific ERP systems. This was a crucial element in

Point to Point Connections	External Central Database	Net-Centric Service
+ Limited parties involved (A&B) and therefore easy to attain initial agreement between them. -Private and proprietary solution for A&B -No transparency for C, D, etc. -Difficult to scale outside A and B -No value chain traceability -Low flexibility to add new content -Complex and difficult to achieve transparency and then only through manual process	+ Data quality + Multiple Data Access Channels -Lack of trust by parties to leave their information outside their control -Difficult to scale as all parties have multiple business relationships -Scalability of the database	+ Information consumption focus + No change in power balance + Scalability + geoscope + sectors mixed + number of companies + depth of information/properties + Control of own data + Individualized data access control + Multiple data access channels - New approach, lack of awareness and knowledge

Figure 7.
Comparison of different architectures

Note: Advantages marked with “+”, disadvantages with “-”

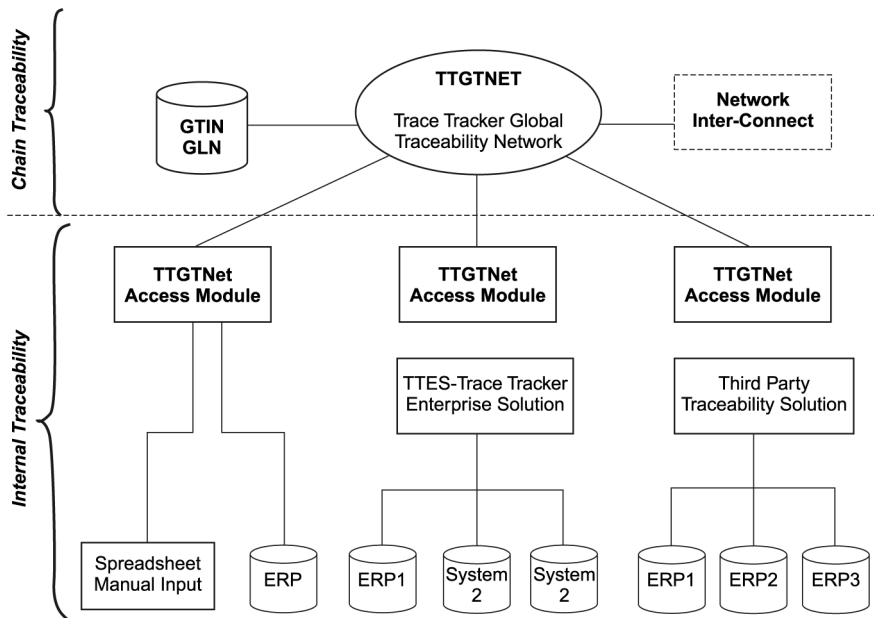


Figure 8.
Implementation
architecture

demonstrating that a net-centric implementation of chain traceability is feasible in a value chain with different internal ERP systems.

The TTES allows each company to visualize the internal traceability for each trade unit or batch and to view relevant properties for each trade unit or batch. The relevant properties are based on the TRACEFISH standards (CEN, 2003a, b).

The net-centric service used for exchanging chain traceability data in the project is called GTNet (Kjærnsrød and Arisland, 2005). Data to the GTNet are delivered through an XML file. The GTNet provides companies with access to relevant data. The access rights are configurable depending for example on ownership of data and agreements between companies.

Results

The evaluation of the implementation study shows several interesting results:

- It has been shown that the TraceFish standards can be used as a basis for the implementation of an electronic chain traceability solution.
- A net-centric approach to an electronic chain traceability solution has been demonstrated.
- Several criteria affecting the implementation readiness of a company have been identified, and the coherence between this readiness and the actual level of implementation is shown.

The results are presented in further detail with emphasis on implementation readiness.

Implementation readiness

A summary of the number of IRC criteria present in the chain is shown in Figure 9.

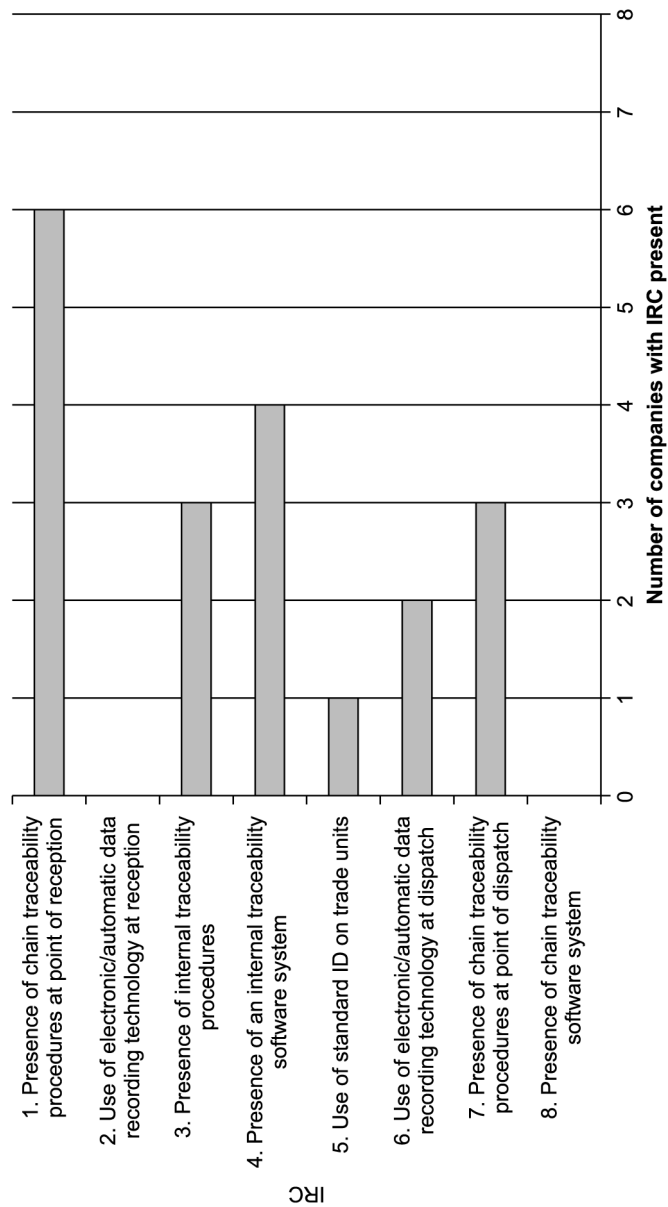


Figure 9.
Number of companies
with the respective IRC
present

The findings presented in Figure 9 show that no chain traceability software system was in use at the start of the project. Automatic data recording was only used at dispatch by two companies. Of the essential TraceFish standard criteria 1, 3, 5, and 7 only criterion 1 “Presence of chain traceability procedures at point of reception” was present in more than 50 per cent of the companies. Criterion 5 “Use of standard ID on trade units” was in use by only one company. Four of the eight participating companies in the value chain had an IT system for handling internal traceability at the start of the project.

When comparing the total number of IRC present in each company with the estimated maximum number of IRC we observed the following results: Two companies have a total number of IRC of 50 per cent or more compared to estimated maximum number (see Figure 10). Company 2 has the maximum score of 75 per cent. Two companies had a score of 12 per cent.

Level of implementation

Complete implementation of the TELOP Trace software concept with online exchange of data was made in the companies 1 and 2. From company 3 and 4 there was not implemented an online exchange of information. Files with extracts from internal databases were transferred to the GTNet as a demonstration. Traceability information from company 6, 7 and 8 was extracted from paper files and transferred to the GTNet as a demonstration. From company 5 no information was exchanged. The levels of implementation are presented in Figure 11, and compared with the corresponding Implementation Readiness Ratio (IRR). IRR is defined by the equation:

$$\frac{\text{IRC found in survey}}{\text{Estimated maximum IRC}} = \text{IRR}$$

Figure 11 shows coherence between IRR and Level of implementation. However, company 1 achieved a full implementation due to management commitment and strategic partnership with company 2.

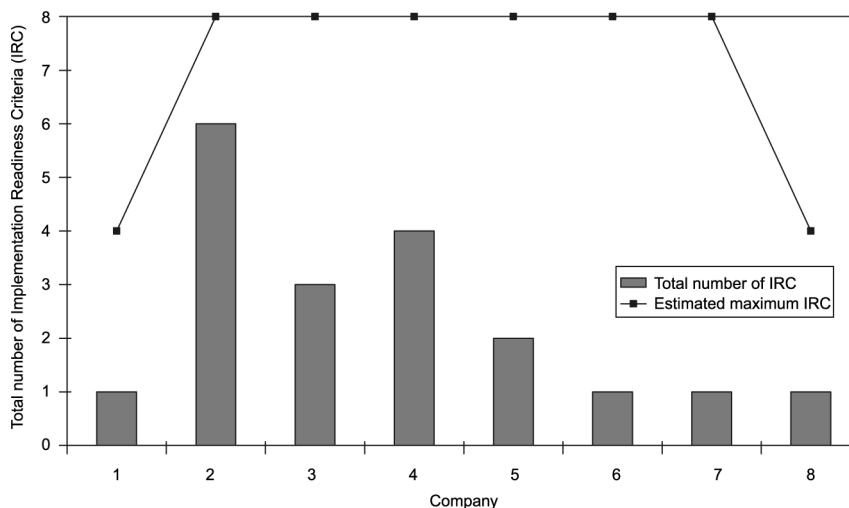
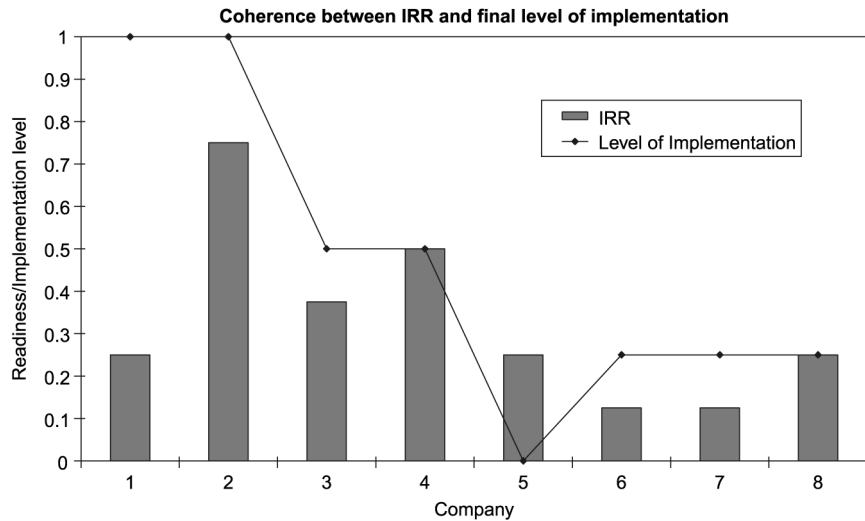


Figure 10.
Implementation readiness
criteria discovered in the
traceability survey

Figure 11.
Implementation readiness
discovered in the
traceability survey



Demonstration – implementation

Full online implementation of the TELOP Trace software concept was made in two companies, 1 and 2. From company 3 and 4 there was not implemented an online exchange of information. Files with extracts from internal databases were transferred to the GTNet. Traceability information from companies 6, 7 and 8 were extracted from paper files and transferred to the GTNet. From company 5 no information was exchanged.

The implementation study showed that the chances of a successful implementation of an electronic chain traceability system depend on a number of factors. In addition to the implementation readiness criteria (IRC) defined in a previous section, strong management commitment is of critical importance. This commitment ensures the allocation of resources to such tasks as changes in internal processes, necessary investments in hardware and software, training of personnel etc. It is also quite common that internal information technology resources and external software providers have a high workload, and there is a constant need for prioritising between important tasks. This also implies that the need for changes in existing business systems will strongly influence a company's ability to implement traceability systems.

Discussion

The Institute of Internal Auditors (Swanson, 2004) states that a conversion to a new system is one of the highest risks that an organization can face. One of their recommended steps is to complete an implementation readiness assessment. Such an audit should develop a detailed understanding of the operational changes that are being planned and facilitate an independent risk assessment of those changes. The TELOP Trace project defines specific criteria (IRC). These criteria are partly based on the Tracefish standard (CEN, 2003a, b) and can be used in an implementation readiness assessment regarding electronic chain traceability. The statistical material in this project is too small to make a conclusion on the correlation between number of IRC and the level of implementation. However, the awareness of the IRC status should be

regarded as an important instrument to detect potential challenging criteria. It is also expected that implementation readiness regarding electronic chain traceability will be an increasingly important criterion for choosing supply chain partners. This will be substantiated by company strategy and management awareness of the importance of electronic chain traceability (Alvarez *et al.*, 2006). This can significantly reduce the risks involved for the collaborating companies.

Electronic supply chain traceability can be seen as an extension of the “Enable Processes” in each company in the chain (Figure 1). In (CSCMP, 2003), some of the important aspects of chain traceability are described as best-practice processes. The use of item identification standards, data standards and transaction standards are of particular importance. Internal traceability systems are used in only four of the eight pilot companies, and seven of the eight pilot companies do not use unique item identification. Being a prerequisite to electronic traceability (Bechini *et al.*, 2005) this was regarded as one of the largest obstacles in the implementation.

As the TraceFish standards are developed for a specific food industry segment, further work on generic food industry standards for electronic exchange of traceability information is needed. This is the focus of a current EU project aiming to improve traceability of food products (TRACE). It is expected that such standardization work will significantly reduce the risks and amount of work involved in implementations.

The organizational implications of electronic supply chain traceability concepts also need to be investigated further. Access rights to traceability data are among the important elements in this area of a supply chain relationship: How can long term storage of traceability data be guaranteed, what happens if a supply chain partner is replaced or files for bankruptcy?

Finally, the study in this article is limited to evaluating the experiences from the implementation of a concept for electronic chain traceability. It will still be necessary to evaluate the long-term operation of such solutions. Important factors will be cost/benefits, organizational impacts, the need for process reengineering and the usability of standards.

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